

Generation Interconnection Request System Impact Study Report

For Queue Project Q14-0359

West Waynesboro 138 kV
April 15, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0359. The Interconnection Customer proposes a 166.2 MW solar photovoltaic facility interconnecting at the West Waynesboro 138 kV substation in the Pennsylvania area. The request was received on July 31, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 178 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$3,412,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
FOUR HILLS to RUBIDOUX 138.0 kV	DRY GAP to WEST KELSO 138.0 kV L	133.1	281	6.8
ALTAVISTA POWER STATION to RUBIDOUX 138.0 kV	WEST KELSO to DRY GAP 138.0 kV	129.0	309	6.8
WATFORD CITY to HUTCHINSON PLANT 1 SUBSTATION 138.0 kV	MORGAN FALLS to FOUR HILLS 138.0 kV	113.7	373	6.9
SUNRISE CTR SUBSTATION to LUTESVILLE 345.0 kV	SUNRISE CENTER to CAYUCOS 500.0 kV line	102.4	1,202	6.9

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
FOUR HILLS to RUBIDOUX 138.0 kV	Rebuild line with higher-capacity conductor	\$9,045,000	22% (\$2,001,000)
ALTAVISTA POWER STATION to RUBIDOUX 138.0 kV	Rebuild line with higher-capacity conductor	\$2,392,000	22% (\$529,000)
WATFORD CITY to HUTCHINSON PLANT 1 SUBSTATION 138.0 kV	Rebuild line with higher-capacity conductor	\$1,850,000	24% (\$440,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
SUNRISE CTR SUBSTATION to LUTESVILLE 345.0 kV	Reconductor line	\$11,128,000	4% (\$442,000)
	Total allocated to project		\$3,412,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.